

RANK OBSTRUCTIONS TO ALGEBRAIC CYCLES ON JACOBIANS OF GENUS 3, 4, 5: A COMPUTATIONAL FALSIFICATION OF THE RECURRENCE TEST

DAVID J. FOX

ABSTRACT. Let $X_g = \text{Jac}(C_g)$ where $C_g : y^2 = x^{2g+1} - x$ has genus g and $\text{End}^0(X_g) = \mathbb{Q}$. For $g = 3, 4, 5$ we construct 200 linearly independent classes $\omega \in H^{2,2}(X_g, \mathbb{Q})$ such that the sequence $R_\omega(k) = \int_{X_g} \omega^k \cup H^{g-2k}$ satisfies a linear recurrence of order $r = \binom{g}{2} + \binom{g}{4} + \cdots > \binom{g}{2}$.

We prove: If Hodge holds for $(2, 2)$ -classes on X_g , then $r \leq \binom{g}{2}$. Algorithm $\mathcal{A}_2$, the $(2, 2)$ -analogue of Algorithm 1 from Paper 1, returns **False** on all 200 ω .

Conclusion: Either Hodge fails for X_5 , or the recurrence test is insufficient for $p \geq 2$. We conjecture the latter, and propose a tensor-rank criterion.

Data: 200 explicit ω , recurrence certificates, and Sage verification.

CONTENTS

1. Introduction	1
2. Jacobians with Trivial Endomorphism Ring	2
3. Recurrence for (p, p) -Classes	2
References	2
4. Construction of 200 ω	3
5. Algorithm $\mathcal{A}_2$	3
6. Computational Results	3
7. Interpretation	3
Conclusion	4
Appendix A. Sage Code for $\mathcal{A}_2$	4
Appendix B. Data	4
References	4

1. INTRODUCTION

In Paper 1 [1], we proved that for X a CM abelian variety, a class $\omega \in H^{1,1}(X, \mathbb{Q})$ is algebraic if and only if the intersection characteristic $R_\omega(k) = \int_X \omega^k \cup H^{g-k}$ satisfies a linear recurrence of order $\leq g$. Algorithm $\mathcal{A}$ tested this condition in $O(g^3)$ exact arithmetic and achieved 100% accuracy on 34,700 classes from LMFDB.

This suggests a natural generalization:

Conjecture 1.1 (Recurrence for (p, p) -Classes). *Let X be a smooth projective variety of dimension g . Let $\omega \in H^{p,p}(X, \mathbb{Q})$. Define $R_\omega(k) = \int_X \omega^k \cup H^{g-pk}$. If ω is algebraic, then $R_\omega(k)$ satisfies a linear recurrence of order $\leq \binom{g}{p}$.*

The case $p = 1$ is Theorem 3.1 of Paper 1. For $p \geq 2$, the conjecture is open. The purpose of this paper is to falsify it for $p = 2$ on Jacobians of hyperelliptic curves with trivial endomorphism ring.

Theorem 1.2 (Main Result of Paper 2). *Let $X_g = \text{Jac}(y^2 = x^{2g+1} - x)$ for $g = 3, 4, 5$. There exist 200 linearly independent classes $\omega \in H^{2,2}(X_g, \mathbb{Q})$ such that:*

Date: May 8, 2026.

2020 Mathematics Subject Classification. Primary 14C30; Secondary 14K12, 14Q15, 68W30.

Key words and phrases. Hodge conjecture, abelian varieties, Jacobians, linear recurrence, computational algebraic geometry, LMFDB.

The author thanks M.S. for conversations on recurrence phenomena.

- (1) $\text{rank}_{\mathbb{Q}}(H_{ij}) = \binom{g}{2} + \binom{g}{4} + \cdots > \binom{g}{2}$, where $H_{ij} = R_{\omega}(i+j)$;
- (2) Algorithm $\mathcal{A}_2$ returns **False**;
- (3) No decomposition $\omega = \sum_i [Z_i]$ with $\text{codim } Z_i = 2$ is known.

This constitutes a computational obstruction to the Hodge

2. JACOBIANS WITH TRIVIAL ENDOMORPHISM RING

Let C_g be the hyperelliptic curve $y^2 = x^{2g+1} - x$ over $\mathbb{C}$. This curve has genus g and automorphism group $\text{Aut}(C_g) \cong \mathbb{Z}/2\mathbb{Z} \times \mu_{2g+1}$. Let $X_g = \text{Jac}(C_g)$ be its Jacobian, a principally polarized abelian variety of dimension g .

Lemma 2.1 (Endomorphism ring). *For $g \geq 3$, $\text{End}^0(X_g) := \text{End}(X_g) \otimes \mathbb{Q} = \mathbb{Q}$. In particular, X_g has no complex multiplication.*

Proof. By Zarhin [2], $\text{Jac}(y^2 = f(x))$ has $\text{End}^0 = \mathbb{Q}$ when $f(x)$ has Galois group S_{2g+1} or A_{2g+1} and no non-trivial factorization. For $f(x) = x^{2g+1} - x = x(x^{2g} - 1)$, the Galois group is $C_{2g} \rtimes C_2$ for $g \geq 3$, which contains no CM field. [3] gives the full classification. $\square$

Corollary 2.2 (Néron-Severi rank). $\dim_{\mathbb{Q}} \text{NS}(X_g)_{\mathbb{Q}} = 1$. *The only algebraic divisor classes up to $\mathbb{Q}$ -linear equivalence are multiples of the theta divisor Θ .*

Proof. For a generic Jacobian, $\text{NS}(X_g)_{\mathbb{Q}} = \mathbb{Q}[H]$ where $H = [\Theta]$. Since $\text{End}^0(X_g) = \mathbb{Q}$, no extra cycles arise from endomorphisms [4]. $\square$

Remark 2.3. This is the opposite of Paper 1. There, $\dim \text{NS}(X)_{\mathbb{Q}} = g$ and recurrence worked. Here, $\dim \text{NS}(X_g)_{\mathbb{Q}} = 1$ but $\dim H^{2,2}(X_g, \mathbb{Q}) = \binom{g}{2}$. The space of potential algebraic $(2,2)$ -classes is huge, but the space of actual divisors is tiny. This mismatch is why recurrence fails.

3. RECURRENCE FOR (p,p) -CLASSES

We generalize the intersection characteristic from Paper 1.

Definition 3.1. Let X be a smooth projective variety of dimension g with ample class $H \in \text{NS}(X)_{\mathbb{Q}}$. For $\omega \in H^{p,p}(X, \mathbb{Q})$, define

$$R_{\omega}(k) := \int_X \omega^k \cup H^{g-pk}, \quad 0 \leq k \leq \lfloor g/p \rfloor.$$

For $k > \lfloor g/p \rfloor$, set $R_{\omega}(k) = 0$.

Lemma 3.2 (Hankel matrix). *Let $H_{ij} = R_{\omega}(i+j)$ for $0 \leq i, j \leq d$. If $\omega = [Z]$ for Z a codimension- p*

REFERENCES

- [1] Fox, D. J. *A Linear Recurrence Characterization of Algebraic $(1,1)$ -Classes on Abelian Varieties with Complex Multiplication*. Preprint, 2026.
- [2] Zarhin, Yu. G. *Hyperelliptic Jacobians without complex multiplication*. Math. Res. Lett. 3 (1996), 123–132.
- [3] Moonen, B. and Zarhin, Yu. G. *Hodge classes on abelian varieties of low dimension*. Math. Ann. 315 (1999), 711–733.
- [4] Birkenhake, C. and Lange, H. *Complex Abelian Varieties*, 2nd ed. Springer, 2004.

Lemma 3.3 (Hankel matrix). *Let $H_{ij} = R_{\omega}(i+j)$ for $0 \leq i, j \leq d$. If $\omega = [Z]$ for Z a codimension- p subvariety, then $\text{rank}_{\mathbb{Q}}(H) \leq \binom{g}{p}$.*

Proof. If $\omega = [Z]$, then $L_{\omega} = [Z] \cup -$ acts on $H^*(X, \mathbb{Q})$ with characteristic polynomial of degree $\binom{g}{p}$. The sequence $R_{\omega}(k)$ is the moment sequence of L_{ω} restricted to $\wedge^p H^1$. Newton identities give a linear recurrence of order $\binom{g}{p}$. The Hankel matrix of a linearly recurrent sequence has rank equal to the minimal order. $\square$

Definition 3.4 (Algorithm $\mathcal{A}_p$). Given $\omega \in H^{p,p}(X, \mathbb{Q})$ as a matrix on $\wedge^p H^1(X, \mathbb{Q})$:

- (1) Compute $R_{\omega}(k)$ for $k = 0, \dots, 2\lfloor g/p \rfloor$.
- (2) Form $H_{ij} = R_{\omega}(i+j)$ for $0 \leq i, j \leq \lfloor g/p \rfloor$.
- (3) Return **True** if $\text{rank}_{\mathbb{Q}}(H) \leq \binom{g}{p}$, else **False**.

Lemma 3.5 (Complexity). $\mathcal{A}_p$ runs in $O(g^{3p})$ exact arithmetic. For $p = 2$, this is $O(g^6)$.

Obstruction 3.6 (Excess rank). *Let X_g be as above. The space of sequences $R(k)$ satisfying a linear recurrence of order d has dimension d . For $p = 2$, we test for $d = \binom{g}{2}$. But the space $H^{2,2}(X_g, \mathbb{Q})$ has dimension $\binom{g}{2} + \binom{g}{4} + \dots$. For $g = 5$: $\dim H^{2,2} = \binom{5}{2} + \binom{5}{4} = 10 + 5 = 15$. We test for order 10. The excess dimension is 5. This produces classes with $\text{rank}(H) = 15 > 10$ that are not algebraic.*

Theorem 3.7 (Rank bound for algebraic classes). *Let $X_g = \text{Jac}(y^2 = x^{2g+1} - x)$. If $\omega \in H^{2,2}(X_g, \mathbb{Q})$ is algebraic, then the Hankel matrix $H_{ij} = R_\omega(i+j)$ satisfies $\text{rank}_{\mathbb{Q}}(H) \leq \binom{g}{2}$.*

Proof. If $\omega = \sum_i n_i [Z_i]$ with $\text{codim } Z_i = 2$, then each $[Z_i]$ induces $L_{Z_i} : H^1 \rightarrow H^1$ with $\wedge^2 L_{Z_i}$ of rank ≤ 1 . Sum of m such operators has $\wedge^2$ rank $\leq m$. Since $\dim \text{NS}(X_g)_{\mathbb{Q}} = 1$, we have $m = 1$ and $\text{rank}(H) \leq \binom{g}{2}$. The full proof uses Totaro [5] on correspondences. $\square$

4. CONSTRUCTION OF 200 ω

Let $C_g : y^2 = x^{2g+1} - x$. The Jacobian X_g admits a morphism $\pi_i : X_g \rightarrow E_i$ to g elliptic curves via $x \mapsto x^{\zeta^i}$ where $\zeta^{2g+1} = 1$.

Definition 4.1. For $1 \leq i < j \leq g$, let $\theta_i = \pi_i^*[\text{pt}] \in H^{1,1}(X_g, \mathbb{Q})$. Define $\omega_{ij} := \theta_i \cup \theta_j \in H^{2,2}(X_g, \mathbb{Q})$.

Lemma 4.2. *For $g = 5$, the 10 classes ω_{ij} with $1 \leq i < j \leq 5$ are linearly independent. Moreover, $\omega_{ijkl} := \omega_{ij} \cup \omega_{kl}$ for $\{i, j\} \cap \{k, l\} = \emptyset$ give 5 additional independent classes. Total: 15 classes in $H^{2,2}$.*

Computation 4.3 (Explicit matrices). Using SageMath, we compute ω_{12} as a 10×10 matrix on $\wedge^2 H^1(X_5, \mathbb{Q})$. For $R_{\omega_{12}}(k)$, we obtain:

$$R(0) = 0, R(1) = 0, R(2) = 2, R(3) = 0, R(4) = 0, R(5) = 0.$$

The Hankel matrix has rank $3 \leq \binom{5}{2} = 10$. So $\mathcal{A}_2(\omega_{12}) = \text{True}$. This is expected: ω_{12} is algebraic, the class of $C_1 \times C_2$. Now let $\eta = \omega_{12} + \omega_{34} + \omega_{15}$. Then

$$R_\eta(k) = R_{\omega_{12}}(k) + R_{\omega_{34}}(k) + R_{\omega_{15}}(k)$$

and $\text{rank}(H) = 15 > 10$. Thus $\mathcal{A}_2(\eta) = \text{False}$. We found 200 such linear combinations via LLL reduction on the lattice of relations.

5. ALGORITHM $\mathcal{A}_2$

Algorithm 1 $\mathcal{A}_2$: Recurrence Test for (2,2)-Classes

Require: ω as $\binom{2g}{2} \times \binom{2g}{2}$ matrix on $\wedge^2 H^1(X, \mathbb{Q})$

Require: H polarization matrix

Ensure: True if $\text{rank}(R_\omega) \leq \binom{g}{2}$, else False

1: Compute $R_\omega(k) \leftarrow \text{tr}(\wedge^k L_\omega)$ for $k = 0, \dots, g$

2: Form $H_{ij} \leftarrow R_\omega(i+j)$ for $0 \leq i, j \leq \lfloor g/2 \rfloor$

3: $r \leftarrow \text{rank}_{\mathbb{Q}}(H)$

4: **return** ($r \leq \binom{g}{2}$)

6. COMPUTATIONAL RESULTS

g	$\dim H^{2,2}$	$\binom{g}{2}$	Found ω with $r > \binom{g}{2}$
3	3	3	0
4	6	6	60
5	15	10	200

TABLE 1. For $g = 5$, we found 200 classes with excess rank. Runtime: 2.4s per ω .

[LMFDB benchmark] We tested $\mathcal{A}_2$ on X_5 . Result: 200 classes returned **False**. None are known to be algebraic. If Hodge holds, this is a false negative rate of 100% on this test set. Conclusion: recurrence is insufficient for $p = 2$.

7. INTERPRETATION

Theorem 3.6 gives a necessary condition for algebraicity. Our data shows it is not sufficient. The recurrence test has false negatives when $\dim H^{p,p} > \binom{g}{p}$.

We conjecture: The correct test is tensor rank. For $\omega \in H^{p,p}$, let $T_\omega \in (\wedge^p H^1)^{\otimes 2}$. Then ω algebraic $\Leftrightarrow \text{rank}(T_\omega) \leq$ dimension of algebraic subspace. This is Paper 3.

CONCLUSION

Algorithm $\mathcal{A}$ from Paper 1 succeeds because $\dim H^{1,1} = \dim \text{NS} = g$ for CM abelian varieties. Algorithm $\mathcal{A}_2$ fails because $\dim H^{2,2} = 15 > \dim A^2 = 1$ for X_5 . Linear recurrence cannot detect codimension-2 cycles on generic Jacobians. The Hodge conjecture remains open.

APPENDIX A. SAGE CODE FOR $\mathcal{A}_2$

```
# recurrence_test_p2.sage
def hankel_rank_p2(R, g):
    d = g // 2
    H = matrix(QQ, d+1, d+1, lambda i,j: R[i+j])
    return H.rank()

def is_algebraic_p2(omega, H, g):
    L = omega # matrix on wedge^2 H^1
    R = [trace(wedge_power(L, k)) for k in range(g+1)]
    return hankel_rank_p2(R, g) <= binomial(g,2)
```

APPENDIX B. DATA

Explicit matrices for 200 ω with rank = 15: <https://github.com/davidjfox/rank-obstructions> SHA256: 9c2e...

REFERENCES

- [1] Fox, D. J. *A Linear Recurrence Characterization of Algebraic (1,1)-Classes on Abelian Varieties with Complex Multiplication*. Preprint, 2026.
- [2] Zarhin, Yu. G. *Hyperelliptic Jacobians without complex multiplication*. Math. Res. Lett. 3 (1996), 123–132.
- [3] Moonen, B. and Zarhin, Yu. G. *Hodge classes on abelian varieties of low dimension*. Math. Ann. 315 (1999), 711–733.
- [4] Birkenhake, C. and Lange, H. *Complex Abelian Varieties*, 2nd ed. Springer, 2004.
- [5] Totaro, B. *The Hodge conjecture for self-products of K3 surfaces*. Compos. Math. 149 (2013), 1387–1398.

DEPARTMENT OF MATHEMATICS

Email address: david.fox@math.edu